Combinations

1

a 120 **b** 17280 **c** 462 **d** 3876

e 21 **f** 35 **g** 10 **h** $\frac{6}{5}$

2 128

3 16 777 216

4

a 4512 **b** 17 296 **c** 21 808

5 211 926

6

a 252 **b** 210 **c** 126

Probability with combinations

7

a $\frac{1}{1287}$ **b** $\frac{25}{1287}$ **c** $\frac{925}{1287}$ **d** $\frac{56}{1287}$

8 $\frac{1}{2}$

9 $\frac{253}{9996}$

10 $\frac{12}{35}$

11

a $\frac{66}{195\,755}$ **b** $\frac{83\,486}{195\,755}$

12

a 15 **b** $\frac{1}{15}$

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13
a 792 **b** $\frac{1}{12}$ **c** $\frac{1}{220}$

Applications

14
a $\frac{4}{15}$ **b** $\frac{22}{65}$

15
a $\frac{91}{204}$ **b** $\frac{113}{204}$

16
a $\frac{4}{143}$ **b** $\frac{49}{143}$

17 $125\,970$

18 $\frac{42}{143}$

19 91

20 6006

21 7350

22 $27\,720$

23
a 15 **b** 63

24 $324\,632$

25 120

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26
a 16 **b** $\frac{1}{16}$ **c** $\frac{1}{16}$ **d** $\frac{1}{4}$

27
a 10 **b** $\frac{2}{5}$

28 20

29
a 165 **b** 95

30
a 20 **b** 6 **c** 30

31
a 126 **b** 35 **c** $\frac{5}{18}$

32
a 20 **b** $\frac{1}{2}$

33
a 150 **b** 310

34 30

35 126

36 672

37 435

38 128

39
a 325 **b** $\frac{1}{325}$

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40

a 1.01% **b** 35.35% **c** 2.02%

41

a 5.52% **b** 0.01% **c** 0.07%

42

a $\dfrac{429}{5168}$ **b** $\dfrac{4739}{5168}$

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